**Kaspar Maria Von Sternberg**

He is the "**Father of Paleobotany**" (1761–1838) was born in Europe. He established the Bohemian National Museum in Prague and is deemed to be the founder of Modern Paleobotany.

**Birbal Sahani**

He is the "**Father of Indian Paleobotany**" (1891–1949). He presented his research on two different areas of Paleobotany (i) The anatomy and morphology of Paleozoic Ferns (ii) The fossil plants of the Indian Gondwana Formations.

## 19.5.1 Fossilization

The process of formation of fossil in the rocks is called fossilization.

Common methods of fossilization includes petrifaction, molds and cast, carbonization, preservation, compression and infiltration.

**Petrifaction**

Minerals like silica slowly penetrate in and replace the original organic tissue and forms a rock like fossil. This method of fossilization can **preserve hard and soft parts**. Most bones and wood fossils are petrified.

**Mold and Cast**

A replica of a plant or animal is **preserved in sedimentary rocks**. When the organism gets buried in sediment it is dissolved by underground water leaving a hollow depression called a mold. It shows the **original shape but does not reveal the internal structure**. Minerals or sediment fill the hollow depression and forms a cast.

**Preservation**

Original remains can be preserved in **ice** or **amber** (tree sap). They protect the organisms from decay. The entire plant or animal is preserved.

**Compression**

When an organism dies, the hard parts of their bodies settle at the **bottom of the sea bed** and are **covered by sediment**. The process of sedimentation goes on continuously and fossils are formed.

**Infiltration or Replacement**

The **precipitation of minerals** takes place which later on infiltrate the cell wall. The process is brought about by several mineral elements such as silica, calcium carbonate and magnesium carbonate. Hard parts are dissolved and replaced by these minerals.




**Living Fossils:** These are living organisms that are similar in appearance to their fossilized distant ancestors and usually have no extinct close features. e.g. *Ginko biloba*.

## 19.5.2 Determination of age of Fossils

The age of fossils is determined by radioactive elements present in it. They may be carbon, uranium, lead or potassium. It is used in paleobotany and anthropology for determining the age of human fossils and manuscripts.

**Radioactive carbon($C^{14}$) dating method**

This method was discovered by **W.F. Libby** (1956). Carbon consumption of animals and plants stops after death and since then, only the decaying process of $C^{14}$ occurs continuously. The time passed since death of a plant or animal can be calculated by measuring the amount of $C^{14}$. present in their body.

### More to Know

**What is the Geologic Time Scale?**

The geological time scale is a system of chronological dating that relates geological rock strata to time, and is used by geologists, paleontologists, and other Earth scientists to describe the timing and relationships of events that have occurred during Earth's history.



10th_Science Unit-19.indd 280 06-12-2021 18:47:14